

MODARTT

Presents

PIANOTEQ 3.6



Welcome

Congratulations on your purchase of **Pianoteq** - the starting point of a brand new generation of pianos, developed from mathematical research done at the Institute of Mathematics of Toulouse at INSA Toulouse, France, and offering unique possibilities to make the piano behave and sound just the way you like.

For support issues and latest news about our products, please visit our website at www.pianoteq.com. If you have any questions or comment, let us know. We always listen to our customers.

Contents

<u>1 Introduction.....</u>	<u>6</u>
1.1 The fourth generation of pianos.....	6
1.2 What makes Pianoteq outstanding.....	7
1.3 Pianoteq versions.....	8
1.4 Features in short.....	9
<u>2 Installing and starting Pianoteq.....</u>	<u>10</u>
2.1 Installation for Windows.....	10
2.2 Installation for Mac OS X.....	10
2.3 Using Pianoteq stand-alone.....	10
2.4 Using VST hosts.....	10
2.5 Hardware requirements.....	11
2.6 Adapt Pianoteq to your hardware.....	11
2.7 Quickly loading fxp, ptq and MIDI files.....	12
2.8 MIDI file player (stand-alone version).....	12
<u>3 Overview.....</u>	<u>13</u>
3.1 Instruments.....	14
3.2 Hints.....	17
3.3 General commands.....	18
<u>4 Tuning section.....</u>	<u>22</u>
4.1 Diapason.....	22
4.2 Temperament.....	22
4.3 Unison tuning.....	23
4.4 Octave stretching.....	23
4.5 Direct sound duration.....	23
<u>5 Voicing section (acoustic).....</u>	<u>24</u>
5.1 Hammer hardness.....	24
5.2 Spectrum profile.....	24
5.3 Hammer noise.....	25

5.4 Strike position.....	25
5.5 Soft pedal.....	25
6 Voicing section (electro-acoustic).....	26
6.1 Pickup symmetry.....	26
6.2 Pickup distance.....	26
7 Voicing section (clavinet).....	27
8 Design section.....	28
8.1 Soundboard.....	28
8.2 String length.....	28
8.3 Sympathetic resonance.....	29
8.4 Duplex scale.....	29
8.5 Quadratic effect.....	29
9 Equalizer and Velocity section.....	30
10 Output section.....	31
10.1 A new acoustic sound radiation model.....	31
10.2 Sound recording mode.....	32
10.3 Compensation.....	33
10.4 Lid position.....	33
10.5 Binaural mode.....	33
10.6 About microphones position.....	34
11 Effects section.....	36
11.1 Action.....	36
11.2 Tremolo.....	37
11.3 Wah-wah.....	37
11.4 Reverberation.....	38
11.5 Limiter.....	38
12 Practice.....	39
12.1 Tutorial 1: tuning.....	39
12.2 Tutorial 2: voicing.....	40
12.3 Tutorial 3: soundboard design.....	41
12.4 Tutorial 4: adapting Pianoteq to your keyboard.....	42

12.5 Tutorial 5: difference between spectrum profile and equalizer	43
<i>13 Pianoteq Pro.....</i>	<i>44</i>
13.1 Note Edit.....	44
13.2 Spectrum profile.....	45
<i>14 Appendix: temperaments construction</i>	<i>49</i>
<i>15 Special acknowledgements.....</i>	<i>50</i>
15.1 Instrument providers.....	50
15.2 Recording engineers.....	50

1 Introduction

1.1 The fourth generation of pianos

The first generation of pianos, starting in 1698 with Cristofori's *pianoforte*, came to maturity at the end of the XIX century with the acoustic grand concert pianos. It was followed in the XX century by electro-acoustic pianos, and then by digital sampled pianos. Now, at the beginning of the XXI century, **Pianoteq** is the first instrument of the fourth generation of pianos, based on a true modelling and offering extraordinary playability and expressiveness.

For the very first time, you can in great detail adapt the piano sound to your own taste. Unique parameters that model the behaviour of real pianos result in great realism, until now only to be experienced on real acoustic pianos.

Using 32-bit internal computation, the calculated piano sound is free from quantization noise. True dynamic timbre results, from the faintest *pianissimo* to the strongest *fortissimo*, for all 127 MIDI velocities available for each note played.

Everything that characterizes a real piano is there: the mechanical noises (optional), the complex sound of pedals and strings in interaction, the percussive impact on *staccato* play (short notes), and of course, most importantly, **the beauty of the piano sound**.

The second generation brought innovative sounds (such as the Fender Rhodes, Wurlitzer, Yamaha CP, and others), whereas the third generation only reproduced recorded samples. Fully modelled, **Pianoteq** allows you to stretch parameters, resulting in new performance styles. **Pianoteq** is thus both a way to emulate existing pianos and an innovative tool for music creation.

Pianoteq is in fact the first **virtual piano factory**: it can produce new brands as well as copies of historical instruments, from harpsichords or pianoforte to more recent electro-acoustic pianos.

Other instruments of the percussions family are progressively added to the **Pianoteq** collection. Visit our website www.pianoteq.com and discover our latest creations.

1.2 What makes Pianoteq outstanding

- It is the first of a new generation of pianos, issued from an outstanding breakthrough technology based on physical modelling, which is foreseen as the technology of the future (patent pending). Notes are really played (“constructed” in real-time, like on a real piano), not just read from the disk or the memory. This explains why **the sound is alive, not static: it is not a simple recording, it is as genuine instrument that responds to the pianist’s finest interpretation.**
- It has been developed in the prestigious *Institute of Mathematics of Toulouse*, at the *Institut National des Sciences Appliquées* in Toulouse, France. Two specialists from this laboratory have been working hard to create this beautiful instrument. Conception of the physical model is from Philippe GUILLAUME, piano tuner, musician, and mathematician whose “Grail” is finding the equations for the “piano soul.” Implementation is from Julien POMMIER, engineer and mathematician, who implemented the physical model to work in real time.
- It is light: it does not require storing huge data involving a whole collection of DVD’s, since it only needs an up-to-date CPU for computing all sounds in real time. It loads easily into RAM, and installation is instantaneous. No problem using it on a modern laptop.
- It can handle ultra low latencies with ASIO drivers on a recent CPU.
- All computations are done in 32 bits float precision: that means no quantization noise in the computed sound, even with the softest pianissimo.
- It offers a beautiful collection of stunningly alive instruments: pianos, rare historic harpsichords and pianoforte, electro-acoustic and electric pianos, vibraphones...

1.3 Pianoteq versions

Pianoteq is available in three versions:

- **PIANOTEQ PLAY** version is for musicians who want to plug it in and play without tweaking the physical model. It includes standard features such as velocity curve, EQ, dynamics, action settings, tremolo, wah-wah, etc.
- **PIANOTEQ** (standard) version offers in addition powerful tools to tweak and adapt the physical model as well as the microphones position.
- **PIANOTEQ PRO** version goes even further with its Note Edit adapted for those who require complete freedom over sound shaping and creation.

All versions offer the same sound, instruments and playability but differ in the range of features and settings. The table below summarizes the main differences between the three versions.

	PIANOTEQ PLAY	PIANOTEQ	PIANOTEQ PRO
Grand pianos K1, C3, M3	✓	✓	✓
Commercial add-ons [1]	✓	✓	✓
KiViR instruments (free)	✓	✓	✓
Preset (fxp) loading [2]	✓	✓	✓
Audio effects	✓	✓	✓
Piano model tweaking		✓	✓
Microphones setting		✓	✓
Note Edit			✓

[1] Commercial add-ons are sold separately and can be loaded in any **Pianoteq** version.

[2] In **PIANOTEQ PLAY**, preset loading is limited to parameters that are present in the interface. Presets built with **PIANOTEQ PRO** can be loaded in **PIANOTEQ**.

This manual describes the general features that can be found in **Pianoteq**. Depending on your version, some of them may or may not be present.

1.4 Features in short

Pianoteq is equipped with all of the features you want on a real piano:

- Continuous velocity from *pianissimo* to *fortissimo*, with progressive variation of the timbre: that makes exactly 127 velocities! A sample-based software program would in theory require hundreds of gigabytes for all these velocities
- Complex resonances that only a model can reproduce in all its richness:
 - Sympathetic resonances of all strings, both without and with sustain pedal
 - Duplex scale (the undamped string parts which come into resonance)
 - Damper position effect when key is released (variable overtones damping)
 - Other special effects like *staccato* and sound continuation when pressing down the sustain pedal a short time after key release (re-pedalling)
- Timbre modification of repeated notes, due to the hammer striking strings which are already in motion instead of being still
- Release velocity
- Four pedals:
 - Progressive sustain pedal, allowing the so-called “half pedal”, but also quarter or tenth’s pedals if you want! ¹
 - *Sostenuto* pedal, allowing you to hold some notes after release without pressing down the sustain pedal
 - Harmonic pedal, allowing you to play *staccato* while maintaining the sustain pedal resonance
 - *Una corda* pedal, also called soft pedal, modifying the sound quality or *timbre* by shifting the piano action to the right (on grand pianos)
- Variable lid position
- Natural instrument noises including:
 - Action key release noise
 - Damper noise at key release (for bass note dampers)
 - Sustain pedal noise: pedal velocity dependant “whoosh” produced by the dampers rising altogether from the strings or falling down
- Choice of microphone position and multichannel mixing (up to 5 mics, 5 channels)
- Microtuning and *scala* format files import
- Various effects including equalizer, keyboard velocity setting, volume, sound dynamics which controls the loudness levels between *pianissimo* and *fortissimo*, reverberation with control of reverberation weight, duration and room size, action settings, limiter, tremolo, wah-wah...

¹ If your piano supports a fine enough MIDI progression when using the sustain pedal.

2 Installing and starting Pianoteq

Pianoteq works on computers equipped with Windows XP/Vista/7, Mac OS X 10.4 or later (Universal Binary), or Linux (x86). You can use the stand-alone version or use **Pianoteq** as VST, Audio Units or RTAS instrument. We recommend visiting www.pianoteq.com where you will find the latest information and a detailed FAQ.



2.1 Installation for Windows

Execute installation program file `pianoteq_setup.exe`. Activation is done when launching **Pianoteq** for the first time. Just follow the on-screen instructions. ASIO drivers are necessary for obtaining a low latency. If using a soundcard which is not distributed with its own ASIO drivers, ASIO drivers can be downloaded at www.asio4all.com.

Optimization

Windows XP users can optimize audio performances through *Configuration ► System ► Advanced ► Performance parameters ► Advanced*, and there select *Best performances for background services*.

2.2 Installation for Mac OS X

Click on the **Pianoteq** package and follow the instructions. Activation is done when launching **Pianoteq** for the first time. Just follow the on-screen instructions.

2.3 Using Pianoteq stand-alone

Using **Pianoteq** stand-alone is very simple. Launch **Pianoteq** and specify your audio and MIDI *Devices* settings in the dialogue box: you are ready to play.

2.4 Using VST hosts

Pianoteq can be loaded by any VST host. You will need to specify, inside the VST host, your MIDI device and the driver you are using.

Warning

Most VST hosts save your modifications and reload them when you restart. If you hear some strange sounds, make sure that all parameters are at their default values, check for example if velocity and equalizer curves are correctly defined.

2.5 Hardware requirements

Pianoteq can be used with any MIDI compatible keyboard. We recommend a touch sensitive keyboard, such as one with full weighted keys simulating the hammer response of a real piano. See tutorial page 42 for adapting **Pianoteq** to your keyboard in the best possible way. For half and quarter pedalling, a progressive sustain pedal is required.

The requirements for using **Pianoteq** successfully are:

- Recent CPU (dual/quad core recommended)
- 256 MB RAM
- ASIO compatible sound card and drivers (with Windows)
- Internet connection (however installation on a computer which is not connected to the internet is possible, see our FAQ <http://www.pianoteq.com/faq>)
- Windows XP/Vista/7, Mac OS v. 10.4 or later (Universal Binary), Linux (x86).

2.6 Adapt Pianoteq to your hardware

When you click on the *Options* button, a window containing the following sections appear:



In the *Midi* section you can:

- Assign MIDI controllers to **Pianoteq** parameters (experienced users only – may produce unexpected sounds with MIDI files containing special instructions)
- Unassign all controllers except pedals and Pitch Bend by choosing *Current Midi Mapping: Minimalistic* (recommended for starting).

Pianoteq computes all notes dynamically (that is, in real time). Thus a fast CPU is required. The minimum requirement is a 1.5 GHz CPU. Be aware that bass notes require more computations than treble, because they contain many more overtones. The *Perf* section allows you to select from among the following features.

- The *Internal Sample Rate* affects the internal sampling rate of **Pianoteq**. A lower frequency requires less computation, thus the lower the capacity of your CPU, the lower you should set this sample rate.
- The *Polyphony* is the number of individual sounds (notes, sympathetic resonances...) which are played simultaneously. The lower the capacity of your CPU, the lower you should set the polyphony. Alternatively, you can choose *Auto (Pessimistic or Optimistic)* for an automatic polyphony setup.

- *CPU overload detection.* Under certain circumstances (a slow CPU, very fast music), the CPU may be overloaded by the number of required computations. *CPU overload detection* might be useful for real time processing, for example when 10 seconds of sound require 12 seconds of CPU computations. Some computations are then by-passed.

2.7 Quickly loading fxp, ptq and MIDI files

You can drag *fxp* and *ptq* files (and MIDI files with the stand-alone version) from the file manager, email client, web browser, and drop them onto the **Pianoteq** interface.

2.8 MIDI file player (stand-alone version)

The **Pianoteq** stand-alone version is equipped with a MIDI player which lets you play and record MIDI files.

You can adjust the playback speed by clicking on x1. A menu will appear where you can choose another value or enter any value between 0.1 and 10. Alternatively, you can click on x1 and drag the mouse to get the value you want.



Click on the record button to start recording your performance live.



Save your MIDI file by opening the *File* menu. There you can also export your recording to an audio wav file.

3 Overview

We come now to a brief overview of **Pianoteq**. You will notice that the interface is divided in two panels:

- An **Instrument panel** containing innovative features presented in three sections called *Tuning*, *Voicing*, and *Design*. You can open each section by clicking on the section name.

The three sections from the Instrument panel, when opened



- An **Audio Engineering panel**, which offers you a unique feature available through the *Output* menu: microphones positioning and mixing. The panel also offers basic features like equalizer, velocity curve, volume, pedals, reverberation and other effects.

The Audio Engineering panel



Adjusting any of the parameters is easy. But you can also simply choose your instrument from the instrument menu on the top left.

3.1 Instruments

Our virtual instruments are based on physical modelling — which simulates the way the sound is generated and how it propagates in the space — and a mathematical analysis of the most famous acoustic instruments. We built **original instruments** derived from their unique characteristics, trying to get the best qualities of each.



3.1.1 Acoustic pianos

Pianoteq 3 comes with three beautiful grand pianos that each has its own personality. The Grand K1 combines stunning dynamics, sharp attacks and powerful fortissimos with a very nice woody percussive quality. Its many qualities make it suitable to all styles of music. The Grand C3 has a warm and coloured sound, suitable for classical, romantic or lyrical music.

The Grand M3 has an attack and presence that will suit jazz and rock music. Various perspectives are provided with each instrument: player, recording (usually 4 mics), close mic, binaural... These are only a few among the infinitely many perspectives that you can create by choosing your own recording settings: mics placement and mixing.



3.1.2 Electro-acoustic pianos

Thanks to new models, **Pianoteq** can now also reproduce famous electro-acoustic pianos from the seventies. An **electro-acoustic add-on**, included for demo evaluation, offers two amazing electro-acoustic pianos. Each instrument is provided with several variants. Using the rich interface features, you can customize them to your own taste.



3.1.3 Clavinet

The **clavinet add-on** is modelled after the well-known D6 electric clavichord. Thanks to the powerful technology provided by physical modelling, it offers in contrast to the original instrument a sustain pedal, an extended range from C0 to C6 (original range F0 to E5), a variable mute bar and continuous pick-up mixing.



3.1.4 Vibraphones

Pianoteq can also reproduce the sound of the vibraphone. A **Vibes add-on**, included for demo evaluation, offers two beautiful vibraphones. V-M is a virtual copy of a Musser vibraphone that belongs to the famous French vibraphonist Dany Doriz. V-B is a virtual copy of a Bergerault vibraphone that belongs to the Condorcet studio in Toulouse (France).



3.1.5 Prestigious historical collection

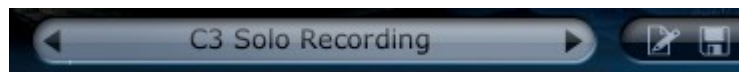
Prestigious historical instruments virtual copies can be downloaded from our website www.pianoteq.com. We currently offer **free add-ons of harpsichords, pianoforte, pianos and electro-acoustic pianos**. Some of these instruments are presented in their current state without modification, including notes that were slightly out of tune or that had voicing irregularities. We keep on enriching this collection.

3.1.6 Other add-ons

Other add-ons are proposed on our website www.pianoteq.com, as for example the **YC5 Rock add-on**. Subscribe to our newsletter for the latest information.

3.1.7 Loading/saving instruments and presets

You can load and save instruments from the *instruments menu*:



This menu lets you choose among a list of built-in *instruments* and *presets*, additional add-on instruments (files with extension .ptq) or presets (extension .fxp) that you have saved in your Pianoteq folders. More details are provided on page 18.

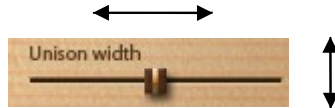
3.1.8 Difference between “instrument” and “preset”

The physical model contains a great number of parameters, but only a small part of them are available through the user interface. When only the user interface parameters are changed, then we say that it is still the **same instrument** but a **different preset**. Hence presets are understood as variations of a given instrument.

For example, *C3 player* and *C3 ambience* are the same instrument, as the difference lies in the microphone position which can be defined by the user himself. On the contrary, *C3 player* and *M3 player* are two different instruments because it is impossible to obtain *M3 player* by modifying the user interface parameters of the *C3 player*.

3.1.9 Instrument modifications

You can change any parameter of the current instrument or preset by moving the corresponding slider with the mouse. Moving the mouse cursor *perpendicularly* to the slider will make the slider move slowly.



Each time you modify parameters in **Pianoteq**, a few computations are needed to update the instrument.

Once you are done, the *edit* and *save* buttons to the right of the *instruments menu* allow you to edit and save your new preset.



Have also fun trying the *Random* button, which changes all settings randomly!

3.2 Hints

3.2.1 Creating your own instrument

Generally speaking, you will find it more interesting to make adjustments, large and small, in several parameters instead of making a single large adjustment. Moving a single slider to the left may throw the weight of the sound onto other parameters that you may want to adjust. Moving a slider to the far right may obscure the contribution other parameters make to the sound, or make their small contribution seem too strong. Moreover, the parameters that interact may be in separate sections, since each section has controls that modify the way in which a single physical component of a piano contributes to the sound, instead of controlling the sound in general. This manual often touches on these adjustments. Further experimentation will let you experience the ways in which the parameters interact. Make a small adjustment in one parameter and a large one in another. Make large and small adjustments everywhere. You can create almost any sound that a piano can create, and more.

3.2.2 Brilliance

Brilliance is an important parameter for achieving good sound realism. You may want to adjust it whether you are staying close to the loudspeakers, or listening at low volume levels, or using headphones. You can for example change the hammer hardness from the *Voicing* section, or use the EQ from the lower panel.

3.2.3 Reverberation

Loudspeakers usually require significantly less reverberation than headphones. The reason is simple: when you use loudspeakers, the sound is naturally reverberated by the room in which they stand.

Hence, you may like to use distinct *reverberation presets* whenever you are listening through loudspeakers or headphones:

- *Player* or *Orchestra* perspective when using loudspeakers,
- *Orchestra* or *Audience* perspective when using headphones.

Moreover, it is worth mentioning that bypassing reverberation in **Pianoteq** does not have the same effect as bypassing reverberation in a sample-based VSTi. In the latter case, there is usually some reverberation present in the recorded samples (unless recording is done in an anechoic chamber), whereas in **Pianoteq**, there is no reverberation whatsoever when you switch it off. In that case, the sound loses an important part of its natural quality and may sound strange or synthetic, particularly with headphones, because in the real world we never hear sounds without some reverberation. Hence, you should bypass reverberation only when using an external (or natural) reverberation.

3.3 General commands

3.3.1 Playing with the parameters

In **Pianoteq** you can modify and create your own presets, save them and share them with other users. The next commands are located on the top of the interface.



- The **edit** (pen) and **save** buttons allow you to edit and save your new preset.
- The parameters **check list** button allows you to select the parameters that you want to keep unchanged when changing instrument or preset. This is a very convenient feature for “transporting” settings from one instrument to another.
- The **random** command allows you to randomly change *tuning*, *voicing*, and *design* parameters. Parameters located in the Audio Engineering panel are not affected by the random command.
- Using the **undo / redo** commands, you can at any time undo and redo the changes you have made: up to 100 undos are possible.
- The **AB** button allows you to switch at any time between two presets A and B, one being in foreground while the other is in background. The *led* right to it is on as soon as A and B are different. Clicking on the led copies the foreground preset into the background preset. A right click on the AB button lists the differences between A and B.

3.3.2 Managing and sorting presets

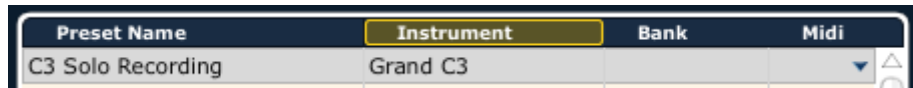
When you click on the edit button, the following presets manager appears:



It lets you manage the instruments and presets from the instruments menu. You can sort them, filter them (see only part of them), rename them, etc.

Sorting

The way presets are sorted can be chosen by clicking in the appropriate column: *Preset name*, *Instrument* or *Bank*.



A *bank* is a set of presets that are located in a given folder, except the *factory* bank which contains built in instruments and add-ons provided as ptq files located in the *Addons* folder.

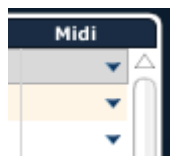
Filtering

You can select the instruments and presets that are visible in the instruments menu through three different “filters” located on the right side of managing window: *search*, *All instruments*, *All banks*. You can also click on individual instruments or banks.



Renaming presets

A double click on the preset name allows you to rename a preset (user defined presets only). Alternatively you can open the contextual menu by clicking on the small arrows in the *Midi* column.



The Pianoteq folders

The easiest way to find the **Pianoteq** folders is to click on the little folder images right to *All instruments* and *All banks*. The **Pianoteq** folder contains two sub-folders:

- *Addons*: this is where to store the add-ons provided as ptq files,
- *Presets*: this is where to store the user defined banks. Each bank is itself a folder that contains presets with fxp file format.

Importing fxp files

There are two ways of importing fxp files:

- Drag and drop the file into the interface,
- Copy the file in the appropriate folder.

Saving presets

After you have created your new preset, you can save it by simply clicking on the save button; a popup window will ask you for a preset name and a bank name (default bank is *My Presets*). Each time you click again on the save button, you will save the new values with the same preset name. You can access to backup copies with a right click on the preset name (if there are some).

3.3.3 Assigning a parameter to a MIDI controller

You can assign **Pianoteq** parameters to MIDI controllers located for example on your keyboard. Select the parameter slider that you want to assign (right click on the slider and click on *Midi Automation*) and move the knob of the MIDI controller that you have chosen for this parameter. Association will be done automatically. The next time you move the MIDI controller, you will see the slider moving in parallel. More sophisticated possibilities are provided under the *Options/Midi* section.

Any MIDI control may be assigned except the following ones: MIDI controls 6, 38, 96-101 which are used for RPN and NRPN MIDI parameters; controls 120 to 127 are reserved for various reset operations ("all notes off" etc.); controls 32 to 63 are reserved for 14-bit precision controllers.

3.3.4 Controls table

The following table summarizes **Pianoteq** main controls that are associated with basic piano features. More details are given in the next sections.

Feature	Controls	Action
Pitch	Diapason	Changes A (above middle C) frequency
Tuning	Temperament	Chooses among standard temperaments
	Unison width	Frequency variation within each unison (group of three strings)
	Octave stretching	Stretches octaves
Sound length	Direct sound duration	Modifies the direct sound duration
	Soundboard mechanical impedance	Modifies the global sound duration: raising impedance yields longer sounds
Timbre	Spectrum profile	Modifies individual intensity of the first eight overtones
	Equalizer	Modifies intensity at chosen control frequencies
	Strike point	Changes the overtones level
	Soundboard cut-off frequency	Raising cut-off frequency enriches high frequencies of each tone
	Soundboard Q factor	Raising Q factor shortens high frequencies duration
	String length	Controls sound “acidity” (inharmonicities)
Brilliance	Hammer hardness	Enriches the high frequencies: the harder the hammer, the brighter the sound
Resonance	Sympathetic resonance	Controls the amount of sympathetic resonances of all strings (the harp)
	Duplex scale resonance	Controls the amount of Duplex scale resonances (undamped string parts)
Reverberation	Wet/dry	Controls the amount of sound reflected by the environment
	Short/long	Controls reverberation duration
	Small/large	Controls main room dimensions
Keyboard	Velocity	Controls the keyboard velocity curve

Warning: when adjusting the parameters to extreme values, one may create instruments having unusual or even unreal properties. Hence it is not surprising that the resulting sound may not correspond to that of any known instrument.

4 Tuning section



Pianoteq allows you to perform all of the tuning operations usually done by a piano tuner. The *Tuning* section contains the following controls:



4.1 Diapason

The standard diapason (A above middle C) frequency² is 440 Hz, but you can change it to values between 415 Hz (-½ tone) and 466 Hz (+½ tone) by clicking on *diapason*.

4.2 Temperament

The *temperament* defines the way the scale is tuned. By clicking on *temperament*, you can choose from among the following temperaments (see appendix):

- *Equal*: the standard tuning
- *Zarlino* (circa 1558): sometimes called the “physicians scale”, based on harmonic thirds (ratio 5/4) and fifths (ratio 3/2)
- *Pythagore* (VIth A.C.): based on harmonic (pure) fifths except for one (the so-called “*quinte du loup*”). Can you hear which one it is?
- *Mesotonic*, *Well-tempered* and *Werckmeister III* (XVIIth): unequal temperaments used in baroque music
- *Flat*: octaves ratio is strictly 2, for use in certain circumstances with Synths, or click on  for importing *scala*³ files and keyboard mappings for *microtuning*.

A unique feature of **Pianoteq** is that tuning does not follow a pre-computed frequency table (except for the flat temperament), but *takes into account the inharmonicity of the strings, in the same way a piano tuner does with acoustic pianos*. Hence, the consonance of the notes is improved and the chords have a fuller and richer sound.

² Frequency is the number of oscillations per second.

³ For more information, see www.xs4all.nl/~huygensf/scala/

4.3 Unison tuning

As very few people know, the three strings of each piano unison (the strings hit by each hammer) are not tuned at exactly the same frequency. To change the *timbre* or *colour* of the sound, a skilled piano tuner introduces small tuning differences between these three strings. Experiment yourself by gently changing

- The *unison width*, that is, the difference between the lowest and the highest frequency produced by the three strings of a single note.

4.4 Octave stretching

It is quite usual to stretch *octaves*⁴ in a piano, but how much should they be stretched? Well..., this might be a matter of taste! Adjust it to your own taste by modifying

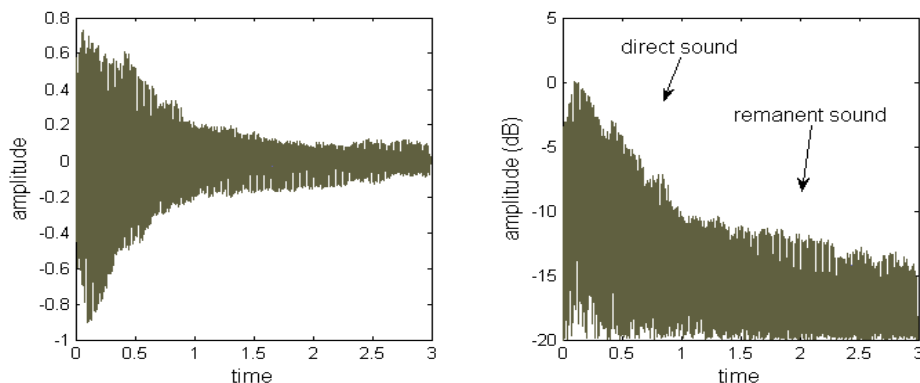
- The *octave stretching* parameter. The main effect will be observed in the treble notes.

When the *octave stretching* parameter is set to 1, the stretching follows the natural inharmonicity of the strings (depending on the string length), so there still is a slight stretching. If you want no stretching at all, then use the *flat* temperament.

4.5 Direct sound duration

One consequence of modifying unison frequencies is that you change the direct sound duration. The same thing happens when reshaping hammers. If you prefer, you can act directly on this duration by changing:

- The *direct sound* duration.



*Time representation of a piano sound decay, natural scale (left) and log scale (right).
Fast decay at the beginning (direct sound), slow decay after (remanent sound).*

⁴ The musical interval between the two closest notes with the same name (e.g. A3 and A4) is called an octave. The theoretical frequency ratio between two such notes is 2, but in practice it is slightly stretched because the ration between the fundamental and the second overtone of a real string is slightly greater than 2 (cf. also Design Section).

5 Voicing section (acoustic)



The strings of a piano are struck by the hammers, small wooden pieces covered with hard felt. For a piano tuner, *voicing* consists in “shaping” the sound according to the pianist’s taste. This is done by working on the hammer felt, giving it the desired shape, hardness and elasticity.

Pianoteq allows you to perform this same voicing operation. The voicing section offers you the following commands:



5.1 Hammer hardness

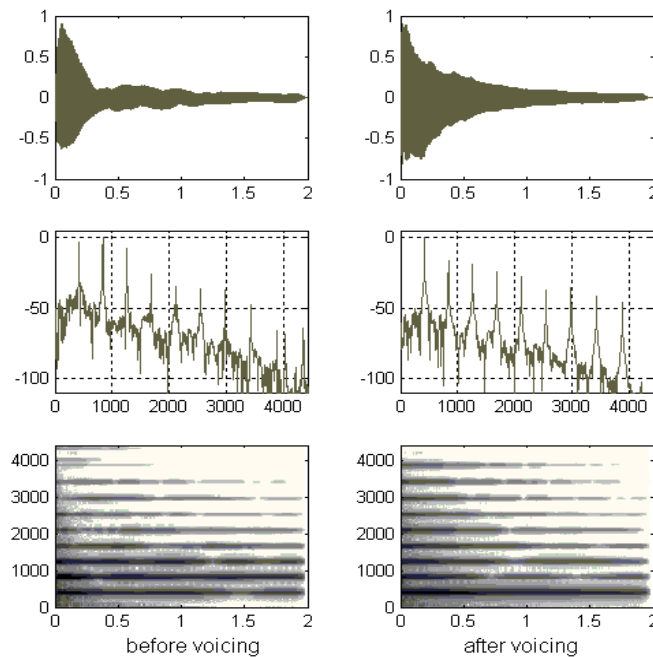
Here you can choose the hammer hardness at three different velocities:

- *Piano*, corresponding to MIDI velocity 32
- *Mezzo forte*, corresponding to MIDI velocity 64
- *Forte*, corresponding to MIDI velocity 96.

The harder the felt, the more brilliant the sound becomes. Of course, the louder you play, the harder the felt should be, unless you wish to try some original sound effects!

5.2 Spectrum profile

Here you will find small sliders that allow you to adjust the individual intensity of the first eight *overtones* (cf. page 28). Experiment with it by increasing the weights of all *fundamentals* (the first overtone is called the fundamental) by raising the first bar. A remark: many people say that piano manufacturers avoid having a strong seventh overtone.



Time, frequency and time-frequency representation of a note, before and after voicing. Here, among other things, the first overtone intensity has been increased.

5.3 Hammer noise

You can adjust the *hammer noise*, that is, the weight of the hammer percussion sound. With a loud hammer noise, you will feel as though you are standing close to the piano.

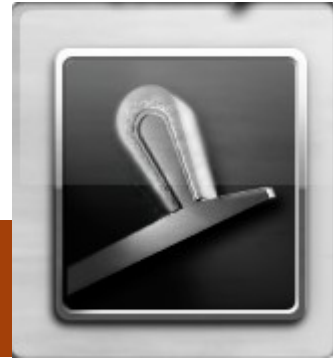
5.4 Strike position

The *strike position* slider allows you to choose at which fraction of the string length the hammer hits the string.

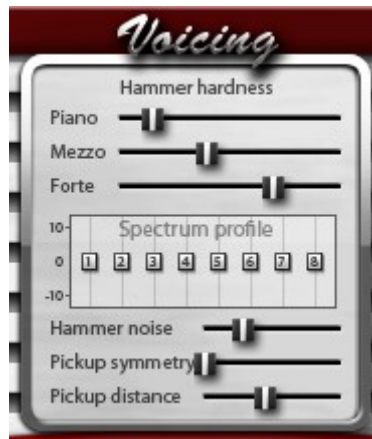
5.5 Soft pedal

Here you can control the smoothing degree of the *una corda* pedal, also called the *soft pedal*.

6 Voicing section (electro-acoustic)



The tone sources (reed, tine, bar) of an electro-acoustic piano are struck by hammers or mallets whose hardness can vary, depending on the material that is used. Here again, as for acoustic pianos, **Pianoteq** provides a voicing section that offers the following parameters:



Hammer hardness, spectrum profile and hammer noise work in the same way as with acoustic instruments.

6.1 Pickup symmetry

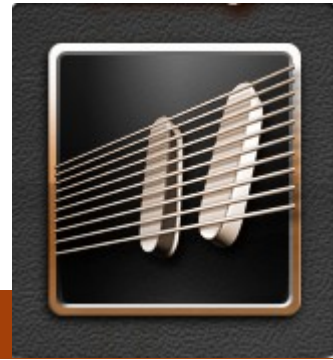
In an electro-acoustic piano, the pickup is not exactly in front of the tone source at rest. When it is exactly in front, due to the symmetry of the device, the note jumps one octave higher than the normal tone. Moving the Pickup symmetry slider from left to right makes the pickup move from an unsymmetrical position to a symmetric position, providing thus a wide range of timbres.



6.2 Pickup distance

The pickup distance slider sets the distance between the tone source and the pickup. When the pickup is moved closer to the tone source, the sound becomes more distorted and the timbre variation between soft and loud sounds increases.

7 Voicing section (clavinet)



The clavinet works similarly to a clavichord. When a key is depressed, a small rubber tip strikes the string and presses it on to an anvil. It contains two sets of pickups, positioned above and below the strings, and is usually electronically amplified.

The voicing section of the clavinet lets you choose among the four standard pickups settings:

- A-C: **L**ower pickup
- B-C: **U**pper pickup
- A-D: both pickups
- B-D: both pickups but out of phase.

Even more, it lets you make any continuous change between these four settings by using the **U**pper pickup and **L**ower pickup sliders.

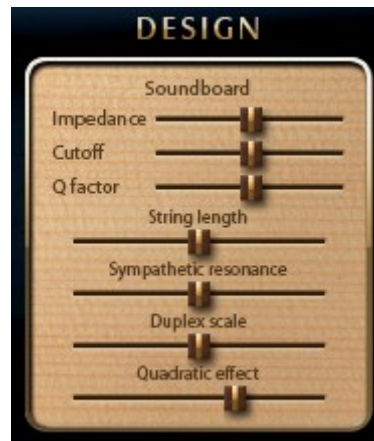


Hammer hardness, spectrum profile and hammer noise work in the same way as with acoustic instruments.

8 Design section



This section offers controls over piano design parameters such as string length or soundboard characteristics (the soundboard is the wooden plate which transmits the string vibrations to air.)



8.1 Soundboard

On a real piano, you cannot change the soundboard mechanical impedance⁵. With **Pianoteq**, it becomes very easy: it is just one of the design parameters. You can control:

- The soundboard mechanical *impedance*: the greater the impedance, the longer the sound becomes.
- The soundboard *cut-off frequency*: the higher you set this frequency, the more high overtones will be present.
- The soundboard *Q factor*: the greater this factor, the faster the high overtones will decrease.

8.2 String length

Each piano note produces a complex sound, mainly composed of *overtones* with approximate frequencies f , $2f$, $3f$... where f denotes the *fundamental* frequency.

A parameter which greatly affects the timbre (and the tuning) is the so-called *inharmonic**ity*: the more inharmonic the strings, the more the overtone frequencies of each string are driven away from their theoretical values f , $2f$, $3f$... and the more the piano sound will resemble a bell.

Inharmonicity decreases very rapidly with string length. Experiment by changing the *String length*. The difference will be most evident in the bass range. You can choose up

⁵ That is, how the soundboard resists the string vibrations, and thus amplifies the sound.

to a 10 meter long piano! At such a size, there is almost no more inharmonicity. People say that piano manufacturers dreamed of producing pianos without inharmonicity...

8.3 Sympathetic resonance

The *Sympathetic resonance* parameter controls the weight of the strings' sympathetic resonances. It is used for example in the famous piece *microcosmos* by Béla Bartók. The sympathetic resonances depend on the position of each individual damper, and consequently on the position of the sustain pedal: it is longer when the sustain pedal is down, as dampers do not touch the strings.

Experiment by pressing down a few notes very slowly so that they do not produce any sound, and then, without pressing down the sustain pedal, play a few notes *staccato*. You will hear the resonance produced by the latter into the first depressed notes. If you release these notes, the sound will stop.

You can also do the “opposite” experiment. Play a note loudly and hold it, press silently another note, release the first note: it continues resonating in the second note.

8.4 Duplex scale

The *Duplex scale* parameter controls the weight of the duplex scale resonance, issued from the undamped string parts located between tuning pins and frame (front scale) and between bridge and frame (rear scale). This invention was patented by Steinway (who was in relationship with the physicist Hermann von Helmholtz) in 1872 and enriches the harmonic content of the notes.

8.5 Quadratic effect

On hard strikes, the nonlinear part of the string response increases, adding new frequencies to the sound. The *quadratic effect* parameter allows you to control the loudness of this nonlinear response.

9 Equalizer and Velocity section



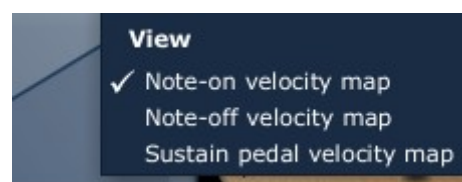
On the left side of the Audio Engineering panel, you can switch between:

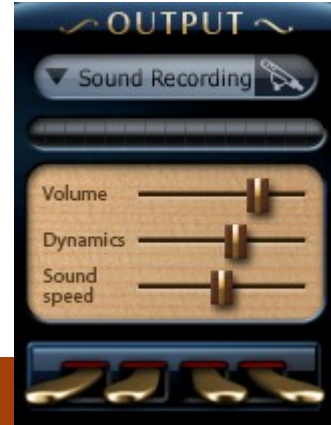
- An *equalizer*, working in **pre-processing** (modifying the model *before* computing sound). The fact that it works in pre-processing allows automatic *volume compensation*: for example, when increasing the medium spectrum, it will increase the medium frequencies in each note while keeping the natural balance across the keyboard, that is, the middle notes will not get suddenly louder than the other ones. Another advantage coming from the pre-processing is that it allows very fine variation in the EQ curve, unlike a standard 3 bands EQ. You can add or remove control points and modify the curve by moving them with the mouse.
- A *velocity* control curve, working on the same principle, allowing you to adjust the response to your keyboard.



A right click on the velocity graphic (or consecutive clicks on the velocity button) opens a menu where you can adjust other types of velocity curves:

- Note-off velocity
- Sustain pedal velocity.





10 Output section

In the middle of the Audio Engineering panel you will find:

- A menu that allows you to choose among four modes: *stereophonic*, *monophonic*, *sound recording* and *binaural*. The *sound recording* mode allows placing **microphones** in the case of acoustic instruments. The *binaural* mode includes a **head model** and is mostly intended for use with headphones. The *stereophonic* and *monophonic* modes are used with electro-acoustic instruments and also allow compatibility with previous versions (Pianoteq 1 and 2).
- The sound *volume*, and the sound *dynamics* which controls the loudness level between *pianissimo* and *fortissimo*. Thanks to the fact that the effect is applied during calculating each note, **dynamics** feature can be seen as the **perfect compressor** as it allows you to adjust the dynamics without any distortion (in particular, attack and decay of individual notes are not modified). A double click on the volume slider allows adjusting the volume note by note.
- The *sound speed*. As this parameter was present all along the model implementation, we chose to make it available here to the user. Its effect is equivalent to changing the time unit. It modifies the travel time of the sound waves in the air.
- Four pedals, from left to right:
 - *Una corda* pedal, also called soft pedal. It moves the piano action to the right, so that the hammers strike the strings differently, changing the *timbre* of the notes
 - Harmonic pedal, allows you to play *staccato* while maintaining the sustain pedal resonance effect
 - *Sostenuto* pedal, allows you to hold some notes after release without pressing down the sustain pedal. Depress some keys, depress the *sostenuto* pedal, release the keys, and the sound of the notes will continue as long as the *sostenuto* pedal is down
 - Progressive sustain pedal, allows so-called “half pedals”, but also quarter or tenth’s pedals if you want! ⁶

10.1 A new acoustic sound radiation model

The **Pianoteq 3** acoustic instruments are based on a new **Acoustic Radiation** model of the sound waves. It simulates the sound field generated by the instrument, and thus allows you to place **microphones** anywhere around the piano, or just above the soundboard, or even below the piano if you want. The sound pressure is measured at the place where you place the microphone, thus simulates a perfect omnidirectional

⁶ If your piano supports a fine enough MIDI progression when using the sustain pedal.

microphone. However, for distant microphones, as the reverb process is decoupled from the sound radiation of the piano, turning the reverb off is equivalent to having a perfect directional microphone oriented towards the piano.



A **head model** is also provided: here, an additional treatment is applied to the sound pressure measured at the ears location of the listener. It provides a high realism when listening with headphones.

10.2 Sound recording mode

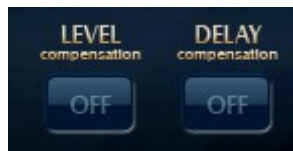
Once you have selected the recording mode, you can choose the microphone position in the *mic area* at the wanted position around the instrument. You can detach the mic window by hovering the mouse cursor above the mic area and drag it with the mouse.

The mix table allows you to mix the microphones into the different output channels. For each active microphone and each active channel, you can adjust the volume and the delay that you want to be applied. Which channels are active depend on your audio configuration.



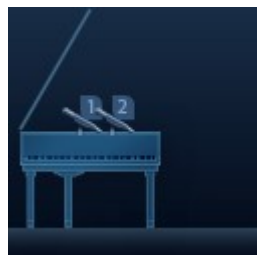
10.3 Compensation

When switched on, *Level* compensation and *Delay* compensation compensate respectively level and delay on each line connecting a microphone to a channel. When switched off, the raw sound given by the microphones is provided: the volume decreases with the distance to the piano while the delay increases. In both cases, level and delay that you adjust in the mix table are relative to the chosen configuration.



10.4 Lid position

The *lid position* can be adjusted from closed to open. This feature is proposed in the same section as the microphone position, because it is part of the acoustic radiation model and thus influences directly the sound waves captured by the microphones.



10.5 Binaural mode

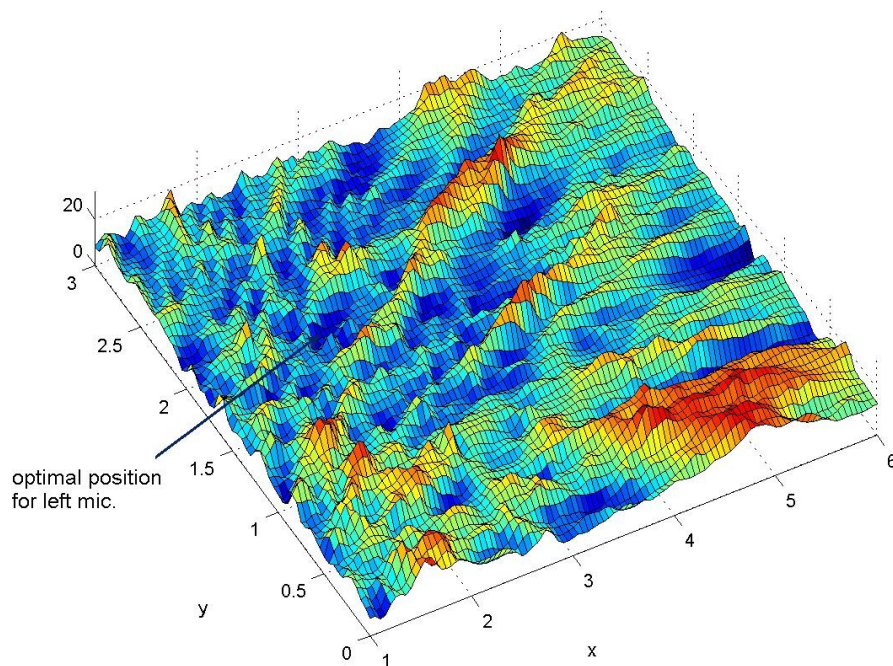
If you have selected the *binaural* mode, you can choose the position of the listener around the piano. You can rotate the head in any direction, and you can even increase its size (for big headed people only).



10.6 About microphones position

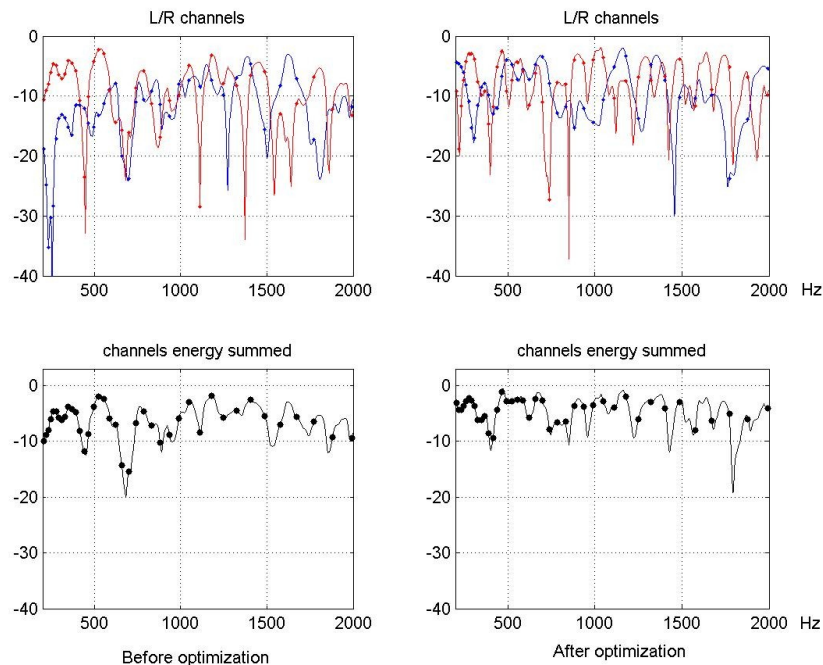
Well-known among sound engineers, the microphone position has a dramatic effect on the resulting sound of the piano: for some positions, the sound is nicely balanced from low to high frequencies while other positions lead to a strongly coloured sound. There might even be places where some frequencies are almost completely missing (case of wave nodes location).

The next graphic gives an idea of how “flatness” can vary with the position: the lower you are on the surface, the flatter the response becomes.



“Flatness” of the global response. The piano is located in the bottom left corner. The microphones are placed to the right of the piano. The graphic is related to the position of one microphone.

The next graphic shows that even the best position is far from rendering a purely flat response: recording always brings some additional colour, which makes sound recording a real art.



Before and after optimization. The dots are located on the frequency of the fundamentals.

In order to make it easier to find good places for the microphones, a quick local optimization of the geometrical parameters of the model (the distances used for the calculation) is performed automatically as soon as you have placed the microphones. This optimization makes the global response of the microphones as flat as possible. As this optimization is done locally (only very small variations of the parameters are allowed), there still are many different kind of sounds to experiment with when moving the microphones.



11 Effects section

In the effects section, you can select among 5 sub-sections: *Action*, *Tremolo*, *Wah-wah*, *Reverberation* and *Limiters*.

11.1 Action

Here you can adjust several parts from the piano action:

- *Damper position*. When a note is released, the damper falls back on the strings and each overtone decays more or less rapidly depending on the damper position. It is particularly noticeable in the bass notes when using partial pedals (and also with keyboards sending variable key release velocity)
- *Damping duration* (efficiency of the dampers)
- *Last damper*: all keys with MIDI note number greater than this value have no damper
- *Mute*: makes the dampers more or less press onto the strings
- *Key release noise*
- *Sustain pedal noise* (“whoosh”) when all dampers rise together, as well as when they fall.



11.2 Tremolo

The *tremolo* is controlled by two parameters:

- The tremolo *depth* in dB which defines the loudness variation
- The tremolo *speed* in Hz.



Besides, the *sync* menu allows you to synchronize the tremolo to the tempo if you wish.

11.3 Wah-wah

The *Wah* slider is the direct controller of the *wah-wah* effect (right click on it and click on *Midi Automation* to assign it to a MIDI controller), its amount being controlled by the *Mix* slider.



In *Auto-Wah* mode, the *wah-wah* effect is triggered by the amplitude of the sound. It is controlled by three parameters:

- The *Sensitivity*. *Auto-Wah* is disabled when the slider is at the extreme left. The more you push it to the right, the lower the amplitude which starts triggering the *wah-wah* effect
- The *LFO rate*
- The *Depth* which controls the frequency shift upper limit of the *wah-wah* filter.

11.4 Reverberation

The *reverberation* is controlled by three parameters:

- The reverberation weight, from *dry* to *wet*
- The reverberation duration, from *short* to *long*
- The room size, from *small* to *large*.



A list of room reverberation *presets* is also provided, with three different perspectives for each preset: *player*, *orchestra*, and *audience*.

11.5 Limiter

By reducing the amplitude from a given threshold, the *limiter* is first intended to avoid casual cracks that appear when the amplitude is too high (in such a case, the clipping led to the right of the vu-meter shows up). It can also be used for producing a distortion effect. It is controlled by three parameters:

- The *Threshold* in dB: the value below which the distortion is less than 3 dB
- The *Sharpness* which makes the limiter active from a large zone to a smaller zone around peaks
- The *Gain* in dB which is applied after the limiter (for special effects).



12 Practice

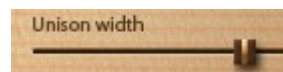
And now, a few tutorials to help you understand how **Pianoteq** works. We will learn how parameters influence the sound. It is important to notice that **not all** parameters settings lead to an acoustic piano sound: the model also lets you produce **new sounds** which could never be obtained from a real instrument.

12.1 Tutorial 1: tuning

What is the difference between a “normal” piano and a “honky tonk” piano? Most of the difference can be found in the unisons tuning. Each note has three strings, except in the bass range. The honky tonk sound comes from the fact that these three strings are not in tune: they do not produce the same frequencies. You can obtain this effect by moving the *unison width* slider to the right in the *Tuning* section:



Normal tuning



Honky tuning

However, having the three strings perfectly in tune is not necessarily best! If you push the slider completely to the left, the three frequencies of each unison will match almost perfectly, but you will find that the sound becomes uninteresting. It sounds too “clean”; it lacks life. So, how should it be tuned? There is no universal truth in such an aesthetic matter. That is why **Pianoteq**, for the first time in a digital piano, lets you adjust the unison tuning to your own taste!

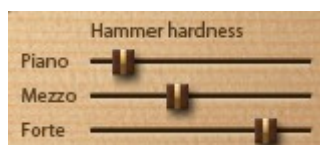
When changing *unison width*, you may also find it interesting to change the *direct sound duration*, reducing it if you have reduced *unison width* and vice versa. Observe that the closer the strings are to being in exact unison, the faster will be the decay of the direct sound and the slower will be the decay of the remanent sound.

Unison tuning is not the only feature that affects tuning: another question is how *intervals*, that is, the frequency ratio between two different notes, are tuned. The tuning of all the intervals within an octave is called a *temperament*.

Over the years, many different temperaments have evolved. The most commonly used today is the equal temperament, in which all semitones are equal. However, a few hundred years ago, people used many other temperaments, some of which you can choose in the *temperament* menu. Try playing them. You may find the difference not so evident when playing single notes, but much more prominent when playing chords, some of them having a nice consonant sound, whereas others will have a quite harsh sound.

12.2 Tutorial 2: voicing

The main objective of piano voicing is setting the brightness of the sound by adjusting the hammer hardness. Different music may require different voicing. The *Voicing* section contains three hammer hardness sliders. Try first moving the *mezzo* slider, which acts on the hammer hardness around MIDI velocity 64. Moving it to the left, you will obtain a softer sound, whereas moving it to the right yields a brighter sound:



Normal setting

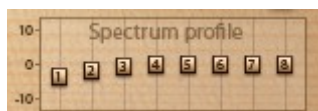


Bright setting

Once you are familiar with these sound changes, you can try the other two sliders acting at *piano* level and *forte* level respectively. You may also want to experiment with using the *hammer noise* parameter in conjunction with this *hammer hardness* setting. You can, for example, set the *hammer hardness* to very soft, but increase the volume of the hammer hitting the string using the *hammer noise setting*. In other words, reducing the hardness of the hammer doesn't mean that you have to lose the percussive sound of the hammer as you reduce the brightness of the sound. On the other hand, you may want to have hard hammers to make the timbre bright, but at the same time reduce the volume of the percussive knock of the hammer hitting the strings.

Another feature is *timbre* adjustment through the intensity of the individual overtones, which can be partially performed on a real piano by shaping or needling the hammers.

- Try moving the overtone sliders one by one in the *spectrum profile*, starting with the first overtone, also called the fundamental.
- Experiment by increasing the eighth overtone by 15 dB. A funny sound, isn't it?
- Next, try some more global shaping, raising or lowering, for example, the three first overtones:



Voicing is not independent from tuning as setting the desired sound length during tuning can be considered as voicing. Do you want to obtain longer decay? Then you can either:

- Act directly on tuning by reducing the unison width in the *Tuning* section,
- Act indirectly on tuning by increasing the direct sound duration,
- Act on design parameter by increasing the soundboard mechanical impedance in the *Design* section.

12.3 Tutorial 3: soundboard design

Soundboard impedance plays a crucial role in piano design. What is the soundboard *mechanical impedance*? When a periodic force is applied to the soundboard, the wood oscillates with a certain velocity at the same frequency as the applied force. The soundboard impedes (resists) with its inertia, elasticity and resistance: this is the mechanical impedance, which has a ratio of *force / velocity*.

Typically, a high impedance results in a long sound because the energy is only slowly transmitted from strings to soundboard, and vice versa. High impedance yields long but weak sounds whereas low impedance yields strong but short sounds. Hence piano manufacturers have to find a compromise between sound level and sound duration.

With **Pianoteq's** solution, you are free to amplify the sound as much as you want. There is no more compromise. You can set the impedance, adapting it to the sound duration that you want:



- The first slider, *impedance*, acts globally on the impedance at all frequencies. Moving it to the right will result in longer sounds and vice versa.

The mechanical impedance depends on the frequency. The impedance is usually quite high below a certain frequency called the *cut-off frequency*, above which it drops down with a slope called the *Q factor*: the rate at which impedance is reduced and thus the rate at which the sound decreases in length. Higher frequencies decay more rapidly than lower frequencies. Hence:

- Moving the second slider, *cutoff*, to the right will increase the cut-off frequency, and thus increase the number of high overtones that are long.
- Moving the third slider, *Q factor*, to the right will increase the impedance slope, and thus decrease the length of the overtones above the cut-off frequency.

Example: if you like a “stringy” sound, try reducing both *Q factor* and *cutoff*.

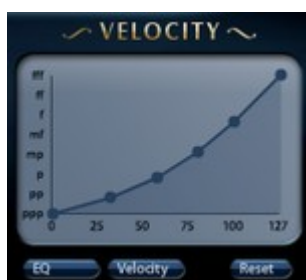
12.4 Tutorial 4: adapting Pianoteq to your keyboard

For demanding pianists, it is of utter importance for an optimal piano experience to adapt **Pianoteq** to the keyboard in the best possible way. The steps that are suggested below are in fact not particular to **Pianoteq**, but could be followed for any other instrument offering the same advanced features.

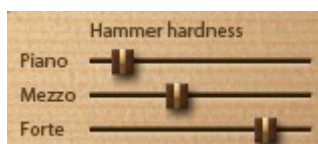
- **Step 1.** If your keyboard has its own velocity settings, choose the one you like the best in general.
- **Step 2.** Select the **dynamics** that you want to use, typically between 30 dB and 60 dB. Measurements done on acoustic pianos show approximately 50 dB dynamics. In piano recordings, due to compression, dynamics can be much less. Concerning volume versus dynamics, we made the arbitrary choice keeping the mezzo forte volume constant when changing the dynamics. So if you want for example to have louder pianissimo while maintaining the same volume for fortissimo, than you need to simultaneously reduce the dynamics and increase the volume.



- **Step 3.** Choose the appropriate velocity curve (right click in the velocity graphic) or build your own. This is an important step, in which you need to set the velocity curve in such a way that you obtain the desired **volume** for each velocity. At this stage, do not take the timbre into account; it comes in the next step.



- **Step 4.** Choose the appropriate **timbre** when playing *piano*, *mezzo forte* and *forte* by moving the corresponding sliders in the Voicing section.



You are now ready to play. Note that steps 2 to 4 can also be used to adapt **Pianoteq** to a given MIDI file in the best possible way.

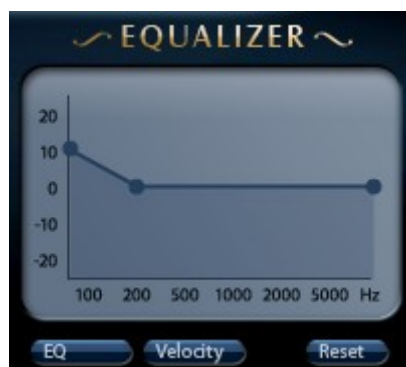
12.5 Tutorial 5: difference between spectrum profile and equalizer

What is the difference between the spectrum profile (see tutorial 2) and the equalizer? Let's look at an example. Open the *Voicing* section and set the first two spectrum bars respectively to +6 dB and +3 dB:



What happens to the sound? **All notes** are modified: for each note, the fundamental has been increased by 6 dB and the second overtone by 3 dB. The piano sound has become globally softer because the higher overtones now have a lower amplitude relative to the fundamental and the second overtone.

Using several *undo*'s, put the spectrum bars back to their original values, and now adjust the equalizer curve, increasing it from 0 dB to 10 dB when going down from 200 Hz to 62 Hz:



What happens now to the sound?

- For all notes with a fundamental **above** 200 Hz, that is, above G2 (the G just below middle C), **nothing** happens, because their overtones are all above 200 Hz, where the equalizer is a straight horizontal line.
- For notes **below** G2, **the first overtones** are modified according to the equalizer curve. For example, G1 whose fundamental is 98 Hz will have its fundamental frequency increased by 6 dB, whereas the other overtones remain almost unchanged (because they are above 196 Hz).

The resulting piano sound will have more bass, with unchanged middle register and treble notes.

13 Pianoteq Pro

Pianoteq Pro is the advanced version of **Pianoteq**. It includes the following additional features:

- Note per note adjustment for physical parameters: tuning, unison width, hammer hardness, strike point, string length, spectrum profile, soundboard impedance, damper position... This note per note adjustment is available in the Standard version only for tuning and volume.
- Sample rate available up to 192 kHz (limited to 48 kHz in the Standard version). Be aware that a high sample rate will generate a massive increase of CPU load.

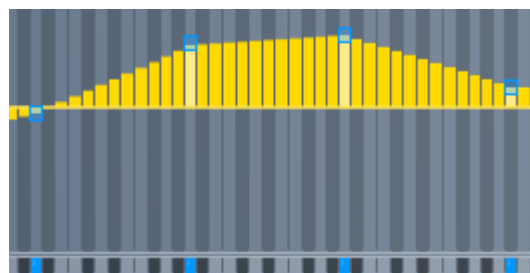
13.1 Note Edit

The *Note Edit* (note per note adjustment) of parameters is a very powerful tool that lets you adjust the chosen parameter for each note.



You access the *Note Edit* window by clicking on the *Note Edit* button or by double clicking on an editable slider. You can drag the Note Edit window at any place and open several new windows, allowing you to work on several parameters at the same time as you can see on the picture above where *Unison width* and *Direct sound duration* are being adjusted.

Observe the *handles* that let you perform some more global changes on the curve. These handles are located on specific *control notes* that you can select/unselect by a double click on the *control rail* below the graphic. These control notes are shared by all parameters except the special case of the spectrum profile which is described in the next section.



Clicking outside a control note allows modifying any particular note. For fine tuning, press the *Shift* key while moving a point of the curve.



Convenient tools are provided that allow you to modify a given shape: *random*, *smooth* and *rescale*. Click and drag the mouse around the button to vary the degree of the effect (see also the tips that are provided on the interface). The *reset* button resets all parameters to the saved settings of the preset (contained in the fpx file on the hard drive).

13.2 Spectrum profile

The Note Edit of the *Spectrum Profile* works a bit differently from the other parameters as you can adjust **each** overtone for **each** note! You can access it by a double click on any of the spectrum profile sliders.

Overtones can be modified individually, but also in different combinations called *harmonic*, *comb*, *major*, *octave*... Various draw modes named *simple draw*, *haircut* and *fill only* are also provided. All these features are grouped in the same menu on the right, which shows *simple draw* at first access.

The spectrum profile has its own set of *control notes* that you can select/unselect by a double click on the *control rail* below the graphic. They are independent of the control notes used by the other parameters.



Outside the control notes, the spectrum profile is linearly interpolated, and you can observe the intermediate values by clicking on the corresponding note on the control rail.

13.2.1 Example 1

If you have one single control note on which you have increased the third overtone by 6 dB, then all notes will have their third overtone increased by 6 dB:



With a single control note...



... all other notes are modified in the same way.

13.2.2 Example 2

Here — starting with a *reset* to clear everything — as we wanted only the middle C to be modified, we added 2 control notes on each side to prevent the interpolation to “propagate” to the other notes:



With three consecutive control notes...



... you can alter only the middle note.

13.2.3 Example 3

In this last example, we modified the middle C and the C below, and we observe how the intermediate notes are interpolated:



Profiling the middle C...



... and the C below ...



... results in interpolated values for the F between.

14 Appendix: temperaments construction

MATLAB programs for some Pianoteq built-in temperaments⁷. Inputs are equal tempered frequencies f (which may be stretched) and corresponding MIDI numbers.

```
%-----
function f = pythagore(f,MIDI)
%-----
delta = 3/2/2^(7/12);
ratio = delta.^[-3 -8 -1 -6 1 -4 3 -2 -7 0 -5 2];
f = f.*ratio(rem(MIDI,12)+1);
%-----
function f = zarlino(f,MIDI)
%-----
ratio2C = [1 25/24 9/8 32/27 5/4 4/3 45/32 3/2 25/16 5/3 16/9 15/8];
ratio = ratio2C./2.^((0:11)/12);
ratio = ratio/ratio(10); % leaving A unchanged
f = f.*ratio(rem(MIDI,12)+1);
%-----
function f = mesotonic(f,MIDI)
%-----
q = 5^(1/4);
ratio2C = [1 5*q^3/16 q^2/2 4*q/5 5/4 2/q 5*q^2/8 q ...
           25/16 q^3/2 4*q^2/5 5*q/4];
ratio = ratio2C./2.^((0:11)/12);
ratio = ratio/ratio(10); % leaving A unchanged
f = f.*ratio(rem(MIDI,12)+1);
%-----
function f = welltempered(f,MIDI)
%-----
q = max(real(roots([1 0 0 2 -8]))); a = (128/q^5)^(1/7);
ratio2C = [1 a^2*q^5/16 q^2/2 a^4*q^5/32 q^4/4 2/a a*q^5/8 q ...
           a^3*q^5/16 q^3/2 4/a^2 a*q^4/4];
ratio = ratio2C./2.^((0:11)/12);
ratio = ratio/ratio(10); % leaving A unchanged
f = f.*ratio(rem(MIDI,12)+1);
%-----
function f = werck(f,MIDI)
%-----
ratio2C = [1 256/243 1.1174 32/27 1.2528 4/3 1024/729 1.4949 ...
           128/81 1.6704 16/9 1.8792];
ratio = ratio2C./2.^((0:11)/12);
ratio = ratio/ratio(10); % leaving A unchanged
f = f.*ratio(rem(MIDI,12)+1);
```

⁷ Some of these temperament definitions are issued from “Der Piano und Flügelbau, Herbert Junghanns, Verlag Das Musikinstrument Frankfurt/Main, 1979”

15 Special acknowledgements

15.1 Instrument providers

- Michel Armengot, France (electric piano CP-80)
- Paul Badura-Skoda, Austria (pianoforte Walter)
- Robin Bigwood, UK (harsichords Grimaldi, Blanchet)
- Olivier Cussac, France (Rhodes MK1, Wurlitzer 200A, Clavinet Hohner D6)
- Luigi Gaggero, Italy (Cimbalom Balázs Kovács)
- Haendelhaus, Germany (pianoforte Schmidt)
- Monique Niebisch, France (grand piano Erard)
- Gilbert O'Sullivan, UK (grand piano Bechstein)
- Daniel Parisot, France (grand pianos Fazioli, Steinway)
- Heinz Preiss, Austria (pianoforte Schöffstoss, Schantz, Graf)

15.2 Recording engineers

- Luc Filaretos (Studio le Graal, France)
- Laurent Minh (Ginkgo, France)
- Herwig Preiss (Austria)

Many thanks also to our customers and distributors for their engaging support and all others who supply us valuable opinions and ideas.

Company

MODARTT is a company that develops and provides software, hardware and consulting services for artistic and technological applications. MODARTT also supports the KIViR project (Keyboard Instruments Virtual Restoration), offering digital restoration of historical keyboard music instruments present in museums. Please visit our website www.pianoteq.com for further details.

Authors

- Pianoteq: Philippe GUILLAUME and Julien POMMIER
- Artwork: Richard BARACETTI
- Manual: Niclas FOGWALL and Philippe GUILLAUME
- Proof-reading: Gregg JOHNSON

Technical data

- Multi channel output (5)
- 32-bits internal computation
- Less than 5 ms latency on a 3 GHz Pentium 4 CPU or equivalent
- 256 voices

Copyright information

- Pianoteq is a trademark of MODARTT S.A.S., 10, avenue de l'Europe, 31520 Ramonville Saint Agne, France
- Cubase, ASIO, and VST are trademark of Steinberg Media Technologies GmbH
- The Audio Units logo is a trademark of Apple Computer, Inc.
- All products names and any trademarks mentioned are used for identification purposes only and are copyrighted by their respective holders.